Python Program for Query Execution

Aim

To write a simple Python program that accepts a query from the user, executes it, and displays the result. This example uses SQLite to run SQL queries.

Algorithm

1. Import the SQLite library.
2. Create an in-memory database connection.
3. Create a sample table and insert records.
4. Ask the user to enter an SQL query.
5. Execute the query and display any returned rows.
6. Close the database connection.

Python Code

```
import sqlite3

# Create an in-memory database
connection = sqlite3.connect(":memory:")
cursor = connection.cursor()

# Create a sample table
cursor.execute(
    "CREATE TABLE students (id INTEGER, name TEXT, mark INTEGER)"
)

# Insert sample records
cursor.executemany(
    "INSERT INTO students VALUES (?, ?, ?)",
    [
        (1, "Arun", 85),
        (2, "Priya", 92),
        (3, "Kumar", 76)
    ]
)

print("Sample table: students (id, name, mark)")
query = input("Enter an SQL query: ")

try:
    cursor.execute(query)

    if cursor.description is not None:
        rows = cursor.fetchall()
        for row in rows:
            print(row)
        if not rows:
            print("No records found.")
    else:
```

```
        connection.commit()
        print("Query executed successfully.")

except sqlite3.Error as error:
    print("Query error:", error)

finally:
    connection.close()
```

Sample Query

```
SELECT name, mark FROM students WHERE mark > 80;
```

Sample Output

```
Sample table: students (id, name, mark)\nEnter an SQL query: SELECT name, mark FROM students WHE
```

Explanation

• **sqlite3** is Python's built-in library for working with SQLite databases.
• **cursor.execute()** runs the SQL query.
• **fetchall()** retrieves all rows returned by a query.
• The **try/except** block displays an error if the SQL query is invalid.

Note

This is a learning example using a temporary in-memory database. Enter SQL statements for the sample **students** table. Only use trusted SQL; this example is not intended as a secure public query service.

Result

The program accepts and executes an SQL query and displays the result or an error message.